AVINABA GHOSH

Software Engineer | Full-Stack Developer | Forward Deployed Engineer
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SUMMARY

Full-Stack Software Engineer with **1+ year of experience** building enterprise and government platforms, cloud applications and **GenAI** solutions directly with clients. Skilled in **Python, JavaScript, TypeScript, React, Next.js, Node.js, SQL** and **Google Cloud Platform**, with hands-on experience in REST APIs, authentication (OAuth 2.0, SSO, RBAC), microservices, workflow automation and LLM integration. **GCP Associate Cloud Engineer** certified.

PROFESSIONAL EXPERIENCE

Cloud Developer & Consultant - BlueVector AI (Full Time) — Apr 2026 – Present

Ohio Department of Youth Services (ODYS) – Government Grants Platform

- Rebuilt a legacy platform into a **full-stack enterprise application** using **React, Next.js, TypeScript, Python, PostgreSQL** and **REST APIs**, with **domain-driven data models** for courts, grants, fiscal periods and allocations across 88 Ohio counties.
- Built **configuration-driven dynamic forms** (schema versioning, conditional logic, validation) and **workflow engines** for grant review, approvals, compliance, document generation, notifications and audit trails.
- Implemented **SSO** and **RBAC** with **OAuth 2.0, OIDC, SAML, Firebase Authentication** and **Google Identity Platform**; deployed on **GCP Cloud Run** (Dev/UAT/Prod).

Professional Licensing Platform & Enterprise AI Assistant

- Developed a **config-driven licensing platform** with dynamic forms, compliance workflows and RBAC using **PostgreSQL (JSONB), pgvector, Pub/Sub, Docker** and **OAuth 2.0** (authorization code, access/refresh tokens).
- Built a **Gemini LLM** pipeline converting unstructured web content into **structured JSON** configurations, **cutting processing time by 70%** (to 15–20s).
- Architected a secure **AI assistant** with **RAG** knowledge retrieval, role-aware access, **Next.js** proxy, **Nginx**, auth guards and API key security.

AI Document & Decision Platforms – State of Michigan (EGLE) & Missouri (MOPBB)

- Built **AI document extraction** and review platforms using **Gemini/Vertex AI, Python, SQL** and **React**, with configurable prompts, extraction profiles and **human-in-the-loop** validation.
- Developed **AI scoring/rubric engines** and role-based dashboards with workflow states (Draft, Submitted, RFI, Approved, Denied).

Healthcare – Lurie (Phase 1 & Phase 2) Telehealth Project & Medicaid Project

- Migrated **Fitbit** integrations to **Google Health APIs** with **OAuth 2.0**, schema transformation and error handling, retrieving **10+ health metrics** (heart rate, SpO_2).
- Built **data pipelines** with minute-wise sync, **NoSQL time-series** storage, threshold alerts and **React dashboards** with RBAC.
- Built a versioned **Medicaid Level-of-Care rule engine** and reusable **React** form components, tested with **Playwright**.

Integrations & Client Delivery

- Built **Asana** integrations in **Python** using **REST APIs, webhooks** and **event-driven** sync for tasks, time tracking and status automation.
- Owned the **full SDLC** with US government and healthcare clients: requirements, POCs, demos, testing, **debugging**, deployment and **production support**.

Associate Cloud Consultant - BlueVector AI (Internship – Converted to Full-Time) — Nov 2025 – Mar 2026

- Built applications, dashboards and workflows using **React, Python, SQL** and **REST APIs**; gathered **client requirements** and delivered **POCs and demos** for US state clients.

PERSONAL PROJECTS

Stories – AI Work Management Platform | Next.js, TypeScript, FastAPI, Python, PostgreSQL, Redis, Celery, WebSockets, Docker

- Built a **multi-tenant** work management platform with Kanban, Gantt and Timeline views, dependencies, critical path, OKRs and timesheets.
- Implemented **real-time collaboration** (WebSockets, Redis Pub/Sub), **Celery** jobs, an automation engine, **RBAC** and **SSO (SAML/OIDC)**.

AI Personal Finance Manager | Next.js, Tailwind CSS, shadcn/ui, Supabase (PostgreSQL), Prisma, Clerk, Inngest, Arcjet, Gemini API

- Built a full-stack expense manager with **Server Actions**, Clerk auth, **Arcjet** rate limiting, receipt **OCR** and **Inngest** background jobs.
- Built a **Gemini AI chatbot** and investment advisor using user transactions, financial health metrics and cached market data APIs.

Query – Q&A Platform | React, Node.js, Express.js, MongoDB, Firebase Authentication, REST APIs

- Built a full-stack **MERN** Q&A platform with Firebase Auth, protected routes, **REST APIs** and question/answer **CRUD**.

SkyWatch Live – Real-Time Flight Tracker | Next.js, React, FastAPI, Python, OpenSky Network API

- Built a **real-time** flight tracker with a **FastAPI** backend processing **OpenSky API** data and an interactive live aircraft map.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, Java, C, HTML5, CSS3
Frontend: React, Next.js, Tailwind CSS, shadcn/ui, Radix UI, TanStack Query, React Hook Form, Zod, Recharts, Responsive Design
Backend: Node.js, Express.js, FastAPI, Pydantic, REST APIs, API Integration, Webhooks, WebSockets, Celery, Background Jobs
Databases: PostgreSQL, MySQL, MongoDB, Redis, pgvector, JSONB, SQLAlchemy, Alembic, Prisma, Data Modeling, Data Migration
Cloud & DevOps: Google Cloud Platform (GCP), Cloud Run, Firebase, Pub/Sub, Docker, Docker Compose, Nginx, MinIO/S3, Git, GitHub
Auth & Security: OAuth 2.0, OpenID Connect (OIDC), SAML, SSO, Firebase Authentication, Google Identity Platform, JWT, RBAC, API Security
AI / GenAI: Generative AI (GenAI), LLMs, Google Gemini API, RAG, Prompt Engineering, Document AI, Information Extraction
Architecture: System Design, Microservices, Multi-Tenancy, Event-Driven Architecture, Config-Driven Architecture, Legacy Modernization
Testing & Tools: Playwright, Pytest, Vitest, React Testing Library, Dialogflow, SendGrid, Agile/Scrum, SDLC, DSA, OOP, DBMS

EDUCATION

B.Tech in Computer Science and Engineering (Future Institute of Engineering and Management) — **Oct 2022 – Jul 2026**
CGPA: **9.08/10 (Rank 1 in college)** | Core Committee: **GDG FIEM** & FIEM Cultural Club: Organised cultural and tech events.
Higher Secondary: **93.33% (2022)** | ICSE: **91% (2020)**

CERTIFICATIONS

GCP Associate Cloud Engineer (Google Cloud) | Machine Learning Training (Internshala) – 85%